Generative AI - Final Report

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January 2026

1 Abstract

The target audience of this project is students. The core problem addressed is the considerable time and effort required to manually create Anki-like flashcards. The primary goal is to support efficient learning by automatically generating structured study cards in the form of question–answer pairs from heterogeneous lecture materials. As an added value, the system reduces uncertainty regarding the relevance of information by identifying key content. This study examines several technical challenges, including robust parsing of heterogeneous input formats, prevention of hallucinated content by restricting generation to source material, filtering of irrelevant or non-academic content and feasibility under limited computational resources.

2 Description

The ideal user journey begins in the Project Overview. The user initiates a new workspace by naming the project and selecting an AI provider, such as OpenAI or LMStudio server. They then configure the generation parameters, specifically the "Thinking Depth" for conceptual complexity and a difficulty level ranging from definition-based (1) to analytical tasks (4).

Following this, the user defines the flashcard scope and density, uploads "Lecture Notes" as the primary source, and adds "Extended Information" for context before clicking "Upload Project" (the process may take a while). Upon completion, the workflow transitions to the Flashcard Overview, where the user can organize, prioritize, or edit the generated deck. Finally, the user enters "Learn" mode to study, rating their confidence on each card to adapt the learning process.

The primary challenges addressed in this work concern the formulation and application of effective prompt engineering strategies. Prompts must be designed in a strict and unambiguous manner to reduce the risk of hallucinations and to ensure that generated content remains consistent with the provided sources. In addition, the decomposition of the overall task into multiple sequential steps plays a crucial role, as the chosen structure directly affects the reliability and consistency of the system outputs.

To address these challenges, the system employs a specialized three-step generation pipeline that separates the overall task into distinct stages. Each stage is handled by an isolated large language model (LLM) persona with task-specific temperature settings. This structured workflow is designed to mitigate hallucinations and redundancy while improving the robustness and dependability of the generated results.

1. **Brainstorming (Extraction) Persona:** Senior Professor (Temp: 0.7)
   Goal: Maximum coverage of "Knowledge Atoms."
   Methodology: The model identifies fundamental facts and generates four stylistic variations for each (Direct, Conceptual, Scenario, and Alternative). The high temperature encourages diverse phrasing to cast a wide "net" over the source material.

2. **Quality Audit (Filtering) Persona:** Strict Auditor (Temp: 0.1)
   Goal: High precision and zero redundancy.
   Methodology: A semantic similarity audit is performed to consolidate overlapping questions. The auditor purges "Admin Fluff" (metadata like author names or dates) and trivial content, using a low temperature to maintain rigid logical consistency.

3. **Refining (Pedagogical Polish) Persona:** Educational Editor (Temp: 0.3)
   Goal: Standalone clarity and difficulty calibration.
   Methodology: Questions are mapped to **Bloom's Taxonomy** ("Remembering" to "Analyzing") to match the target difficulty. This stage strips away "meta-statements" (e.g., "In this slide..."), ensuring the flashcards function as standalone study units.

3 Evaluation

The evaluation of the system is designed to assess the quality, reliability, and efficiency of the generated flashcards. Comparative experiments are conducted to analyze the influence of varying prompting strategies on system performance. The evaluation makes use of quantitative metrics, namely the **Card Acceptance Rate**, the **Hallucination Rate**, and the **Triviality Rate**, to quantify the overall usefulness, factual correctness, and educational relevance of the generated flashcards.

Table 1: Evaluation Results for 1 Steps

	File 1	File 2	File 3
Generated Cards	27	45	36
Card Acceptance Rate	19/27	42/45	36/36
Hallucination Rate	1/27	0/45	1/36
Triviality Rate	5/27	1/45	0/36

Table 2: Evaluation Results for 2 Steps

	File 1	File 2	File 3
Generated Cards	45	45	58
Card Acceptance Rate	35/45	44/45	45/58
Hallucination Rate	0/45	4/45	0/58
Triviality Rate	6/45	0/45	4/58

Table 3: Evaluation Results for 3 Steps

	File 1	File 2	File 3
Generated Cards	27	43	36
Card Acceptance Rate	24/27	38/43	34/36
Hallucination Rate	3/27	2/43	4/36
Triviality Rate	1/27	2/43	0/36

The results indicate that increasing the number of generation steps leads to a higher number of accepted flashcards; however, it also increases the likelihood of hallucinations due to more frequent access to the LLM. At the same time, a multi-step generation process reduces the number of trivial cards, resulting in more meaningful content overall. The system demonstrated reliable performance on the provided material. Nevertheless, limitations in computational resources required a reduction in the number of slides and PDF documents processed simultaneously. Despite this constraint, the selected subset of materials remained representative of the overall dataset.

4 Reflection

The proposed system is designed to increase human agency by ensuring that students remain actively involved in the learning process. Users retain full control over the generated flashcards, including the ability to decide whether a card is relevant and to delete cards that are not considered useful. While the system supports summarization and enhances the understanding of the learning material, it does not replace autonomous decision-making. Students continue to take intentional actions by selecting which cards to study and how to engage with them.

Future development of the system includes extending the parsing capabilities to support handwritten materials and adding references to the generated flashcards that link back to the corresponding slides. Additionally, methods must be developed to extract only relevant pages from supplementary literature to avoid exceeding the maximum allowable number of tokens during processing.